implosion. The high-speed shrinkage of a star produced by the pull of its own gravity.

inertia. A body's resistance to being accelerated by forces that act on it.

inertial reference frame. A reference frame that does not rotate and on which no external forces push or pull. The motion of such a reference frame is driven solely by its own inertia. See also *local inertial reference frame.*

infrared radiation. Electromagnetic waves with wavelength a little longer than light; see Figure P.2 on page 25.

interference. The manner in which two waves, superimposing on each other and adding linearly, reinforce each other when their crests match with crests and troughs with troughs (constructive interference), and cancel each other when crests match up with troughs (destructive interference).

interferometer. A device based on the interference of waves. See *radio interferometer* and *interferometric detector.*

interferometric detector. A detector of gravitational waves in which the waves' tidal forces wiggle masses that hang from wires, and the interference of laser beams is used to monitor the masses' motions. Also called *interferometer.*

interferometry. The process of interfering two or more waves with each other.

intergalactic space. The space between the galaxies.

interstellar space. The space between the stars of our Milky Way galaxy.

inverse square law of gravity. Newton's law of gravity, which says that between every pair of objects in the Universe there acts a gravitational force that pulls the objects toward each other, and the force is proportional to the product of the objects' masses and inversely proportional to the square of the distance between them.

ion. An atom that has lost some of its orbital electrons and therefore has a net positive charge.

ionized gas. Gas in which a large fraction of the atoms have lost orbital electrons.

jet. A beam of gas that carries power from the central engine of a radio galaxy or quasar to a distant, radio-emitting lobe.

laws of physics. Fundamental principles from which one can deduce, by logical and mathematical calculations, how our Universe behaves.